

# RAHUL PRASAD

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Portfolio: <https://rahulprasad-ds.vercel.app/> | GitHub: [github.com/rprasad753-dot](https://github.com/rprasad753-dot)

## PROFESSIONAL SUMMARY

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Machine learning engineer with a mechanical engineering background, trained to trace failures back to their cause and now applying that same habit to data. Builds end-to-end ML pipelines in Python, Scikit-learn, Pandas, and NumPy, from preprocessing and feature engineering through model evaluation and deployment. Experience spans supervised and unsupervised learning for predictive analytics, customer segmentation, and anomaly detection, plus LLM and RAG applications built during a data science internship at OSPYN. Looking for a machine learning or data science role where models have to answer real business questions.

## TECHNICAL SKILLS

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**Programming Languages:** Python

**ML Libraries:** Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn

**ML Techniques:** Supervised/Unsupervised Learning, Regression, Classification, Clustering, Feature Engineering, Dimensionality Reduction

**Algorithms:** Linear Regression, Logistic Regression, Decision Trees, K-Means, DBSCAN, PCA, Random Forest

**Model Evaluation:** Accuracy, Precision, Recall, F1-Score, Confusion Matrix, Cross-Validation, Hyperparameter Tuning

**Data Skills:** Data Preprocessing, Data Cleaning, EDA, Missing Value Imputation, Outlier Detection, Data Visualization

**Core Competencies:** Machine Learning Pipeline Development, Data Analysis and Visualization, Predictive Modeling, Statistical Analysis, Problem Solving, Data-Driven Decision Making, Cross-functional Collaboration, Technical Documentation

## PROFESSIONAL EXPERIENCE

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### Data Science Intern

June 2026 – September 2026

OSPYN | Trivandrum, India

- Built an end-to-end telecom customer churn prediction project in Python, covering EDA, data cleaning, feature engineering, and visualization to pinpoint the factors driving churn.
- Trained and compared Logistic Regression, Decision Tree, Random Forest, SVM, and XG Boost models, scoring each on accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC.
- Developed LLM-powered analytical applications and a multi-document RAG system using APIs, embeddings, vector search, and Streamlit to return data-backed answers.
- Turned raw datasets into insights that supported business intelligence and strategic decisions.
- Worked with cross-functional teams to define data requirements and deliver scalable analytical solutions.

### AI and Machine Learning Trainee

November 2024 – June 2025

GITH Global India Techno Hub | Kerala, India

- Designed, trained, and deployed regression and classification models in Scikit-learn, improving them through feature engineering and hyperparameter tuning.
- Ran EDA on structured and unstructured datasets, handling missing values and outliers to raise data quality before modeling.
- Applied K-Means, DBSCAN, and PCA to uncover patterns in high-dimensional datasets.
- Evaluated model performance with accuracy, precision, recall, F1-score, and confusion matrices.

## **Assistant Technical Designer**

February 2023 – February 2024

*HYZ Mechdrive* | Kerala, India

- Designed and optimized mechanical components in CAD, reducing design errors and improving manufacturing precision.
- Produced technical documentation and engineering drawings that kept design, assembly and production teams aligned.
- Implemented design changes that improved assembly efficiency and reduced production costs.

## **Mechanical Engineer Trainee**

June 2018 – December 2021

*Travancore Titanium Products Limited* | Kerala, India

- Monitored pumps, conveyors, and rotary systems, sustaining 98% uptime with minimal downtime.
- Supported predictive maintenance by analyzing equipment data and writing maintenance reports.
- Traced recurring failure patterns through systematic analysis, improving system reliability.

## **KEY ACHIEVEMENTS**

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- Sustained 98% uptime across pumps, conveyors, and rotary systems at Travancore Titanium Products Limited.
- Delivered an end-to-end telecom churn prediction project at OSPYN, benchmarking five classifiers on ROC-AUC, F1-score, precision, and recall.
- Built a multi-document RAG system with embeddings, vector search, and Stream lit.
- Moved from mechanical engineering into machine learning through an Advanced Diploma in AI and ML plus two hands-on industry roles.

## **EDUCATION**

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### **Advanced Diploma in Artificial Intelligence and Machine Learning**

November 2024 – Jun2025

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### **Bachelor of Engineering in Mechanical Engineering**

2013 – 2017

Noorul Islam Centre for Higher Education